

ANDREY OSIPOV

Python Backend Engineer

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SUMMARY

Python backend engineer with 3+ years building and shipping production services. Specialized in high-load systems on FastAPI and Django — performance optimization, large-scale data (SQL and NoSQL), testing, observability, and CI/CD. Comfortable owning services end to end, from database schema and API design to infrastructure and deployment. Background in information security (MEPhI) — I write code with security and load in mind.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Backend: FastAPI, Django, Django REST Framework, aiohttp, SQLAlchemy, Celery, REST API, OpenAPI/Swagger

Databases: PostgreSQL, MongoDB, Redis, ClickHouse, Apache Cassandra

Messaging / Streaming: Apache Kafka, RabbitMQ

DevOps & Infra: Docker, Docker Compose, Kubernetes, Nginx, CI/CD (GitHub Actions, GitLab CI), Linux

Monitoring: Prometheus, Grafana, GlitchTip

Practices: Pytest, Git, Agile, concurrency (threading, multiprocessing, async)

EXPERIENCE

Safe Transport Innovation Center

Feb 2026 – Present

Python Backend Engineer

Moscow

High-load services for transport-security systems; multi-terabyte MongoDB (largest collection 7.2B records), message brokers, Redis, FastAPI, Docker; heavy parallel data processing; sensitive data under NDA.

- Designed and shipped an aggregating collection over a multi-terabyte MongoDB store, consolidating dozens of disparate collections into a single complete model. Cut time to retrieve a full record summary from 3–5s to <100ms and reduced database load by ~40%, removing the need to gather data across many queries.
- Built a Git-style state-versioning system from scratch (backend + frontend): change commits, state diffs, stable versions, and a graph visualization of history — making changes traceable and surfacing regressions (missing/changed data) between releases instantly. Halved acceptance-testing time.
- Contributed to and maintained a high-load parallel data-processing system (threading, multiprocessing), ensuring stability and correctness under peak load.
- Introduced centralized error monitoring (GlitchTip) and uptime alerting to Telegram across all my services — cut mean time to detect (MTTD) from hours to minutes and eliminated silent failures.
- Systematically modernize legacy services by rewriting them on async FastAPI: reduced p95 latency from ~500ms to 80ms, increased throughput 1.5–2x, and steadily paid down technical debt.
- Work on sensitive systems under NDA with strict security and code-quality requirements — directly aligned with my information-security specialization (MEPhI).

Vendor Hub

Sep 2025 – Jan 2026

Python Developer (Contract)

Remote

Contract build of a B2B vendor-matching platform (vendorfind.ru) for a grant-funded startup. From a spec (Figma mockups + product description) I single-handedly designed, shipped, and maintain the platform — architecture, backend, frontend, infrastructure, and deployment.

- Solo-built and launched a B2B marketplace in 3 months; today it hosts ~50 active companies and a base of 2,000+ vendors. Owned the full cycle from database schema to UI and CI/CD.
- Implemented the product core from scratch — a multi-factor algorithm that matches and ranks vendors against project requirements, automating matching to improve relevance and save users time.
- Built a high-performance backend on FastAPI, SQLAlchemy, and PostgreSQL. Offloaded analytics generation and other heavy operations to background tasks (Celery), keeping API response times consistently under 200ms.
- Integrated external services and automated notifications so key flows (new requests, status changes) run without delay.

- Developed a modern frontend on Nuxt (Vue) with dashboards for both roles, without a dedicated frontend engineer.
- Set up full CI/CD (GitHub Actions, Docker): testing, build, and deploy take minutes, with a fully reproducible environment.

Rutube

Sep 2024 – Aug 2025

Python Backend Engineer

Moscow

Leading Russian video-hosting platform. Django (DRF) monolith plus aiohttp and FastAPI microservices; team of 15 engineers, full ownership of my services.

- Optimized 10+ API endpoints (including high-load ones) using Prometheus/Grafana data: reworked ORM queries (select_related/prefetch, indexes), added Redis caching and rate limiting. Cut p95 latency from ~1s to a stable <100ms under horizontal scaling (~100 RPS per instance).
- Eliminated 10+ minute SQL queries that put critical load on the database: refactored logic + Django CTE dropped execution time to seconds.
- Established testing: moved part of the codebase to pytest (200+ tests), raising coverage from below 50%, with distributed CI/CD runs — a faster pipeline and more reliable releases.
- Migrated API docs to OpenAPI 3.0: replaced the deprecated django-rest-swagger with drf-spectacular and documented 150+ endpoints — clear API contracts for the Go and frontend teams.
- Ran a security audit: closed SQL injections and replaced insecure libraries — at the intersection of backend and my information-security specialization (MEPhI).
- Unified Dockerfiles for two services and adopted the uv package manager — faster builds and smoother onboarding.

NPO Mashinostroyeniya

May 2023 – Aug 2024

Python Developer

Moscow

Corporate web development (online stores, landing pages, informational sites).

- Developed the backend for 4 corporate web projects on Python/Django with PostgreSQL: user flows, catalogs, and admin panels — all shipped to production.
- Automated client-request handling: server-side form processing with PostgreSQL storage and instant notifications — cut business response time from hours to 1–2 minutes.
- Integrated external services — payment systems, web analytics, and email — ensuring correct payment handling and behavioral-data collection.
- Set up production infrastructure: hosting, domains, and SSL (HTTPS) for secure, reliable sites; optimized load and responsiveness (caching, SQL/static optimization, image compression).
- Built responsive interfaces (HTML/CSS/JavaScript) and maintained live projects, extending functionality to business needs and resolving production incidents.

EDUCATION

MSc, Information Security — National Research Nuclear University MEPhI, Moscow

Expected 2027

Faculty of Cybernetics and Information Security

BSc, Information Security — National Research Nuclear University MEPhI, Moscow

2025

LANGUAGES

Russian — Native English — B2 (Upper-Intermediate)